

CAD Importer

From CAD file to simulation-ready Unity prefab in five minutes. Full details: **CAD Importer — Documentation.pdf** (same folder).

1 Install

Window → **Package Manager** → + → *Add package from git URL...*

```
https://github.com/Motawe3/CADImporter.git?path=Packages/com.motawea.cad-importer
```

Append `#v1.0.0` to pin a release. Nothing else in your project is touched. Works with URP (recommended) and the Built-in pipeline.

2 Import your first model

Drop an `.stl` or `.ply` file anywhere under `Assets/`, then drag the resulting prefab into your scene. Done — the asset already contains:

- meters & Y-up conversion (from CAD millimeters, Z-up),
- welded vertices and smooth normals with crisp 30° hard edges,
- a 3-level LOD chain (100% / 50% / 15% triangles) per part,
- a simplified physics collider per part,
- a `CADModelInfo` component with source metadata.

Select the asset in the Project window to tune any of this in the Inspector, then press **Apply**. For many files at once, use **Tools** → **CAD Importer** — queue files, set options once, import in one click.

3 Enable STEP / IGES (optional)

1. Install **FreeCAD** (free) from **freecad.org** — its Open CASCADE kernel performs the B-rep tessellation.
2. Open **Tools** → **CAD Importer**. FreeCAD is normally auto-detected; otherwise point **Browse...** to `FreeCADCmd.exe` (e.g. `C:\Program Files\FreeCAD 1.0\bin`).
3. Drop `.step` / `.iges` files into your project. Every solid becomes a named child GameObject — assemblies keep their structure.

4 Pick the right collider for robotics

Use case	Collider Mode
Static environment: frames, fixtures, cells	Simplified Mesh (default)
Moving robot links on Rigidbody / ArticulationBody	Convex Mesh
Coarse interaction only	Box
Precise contact analysis, static	Full Mesh

5 Load CAD at runtime (digital twins)

```
using CADImporter;

GameObject model = await CADRuntimeImporter.ImportAsync(
    @"C:\twins\gripper.stl",
    new CADRuntimeImportSettings
    {
        sourceUnit          = SourceUnit.Millimeters,
        generateColliders   = true,
        convexColliders     = true    // for dynamic Rigidbodies
    });
```

Parsing and processing run on a worker thread, so the simulation never hitches. Supported at runtime: STL, PLY, OBJ. For a ready-made scene, import the **CAD Importer Demo** sample from the Package Manager (requires URP), or add the `DemoCadRuntimeImporter` component to any GameObject.

Builds: runtime materials use the URP Lit shader — add it to *Project Settings* → *Graphics* → *Always Included Shaders* (or pass your own material in the settings).

Defaults worth knowing

Setting	Default	Change it when...
Source Unit	Millimeters	the file was exported in cm / m / inches
Source Orientation	Z-up right-handed	model imports lying on its side
Smoothing Angle	30°	surfaces look faceted (raise) or edges look mushy (lower)
LOD Qualities	0.5, 0.15	you need more or fewer detail levels
Collider Quality	0.25	contact precision vs. physics cost
Mark Static	Off	importing non-moving factory geometry — turn it on

CAD Importer 1.0.0 · Requires Unity 6000.0+ · MIT License · Support: github.com/Motawe3/CADImporter/issues · Full manual: *CAD Importer — Documentation.pdf*